

Statically indeterminate beams

When equilibrium runs out, compatibility supplies the missing equations

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01 Counting

A planar beam has three independent equilibrium equations: forces along the beam, forces across it, and moments. When the supports provide more reaction components than that, statics alone cannot find them. The surplus is the degree of static indeterminacy. For a single beam, with r reaction components, it is $r - 3$ (Hibbeler, *Structural Analysis*, counts determinacy the same way). When there is no axial load, the horizontal equation and the horizontal reactions both drop out, so the count is unchanged for a propped cantilever or a continuous beam, but a beam fixed at both ends then has two redundants for bending, not three.

— *Equilibrium gives three equations; extra supports add unknowns it cannot find*

Beam	Reactions r	Equations	Degree $r - 3$
Simply supported (pin + roller)	3	3	0
Propped cantilever (fixed + roller)	4	3	1
Two-span continuous (pin + 2 rollers)	4	3	1
Fixed at both ends	6	3	3

Each redundant needs one **compatibility** equation: a deflection or slope the supports force to be zero.

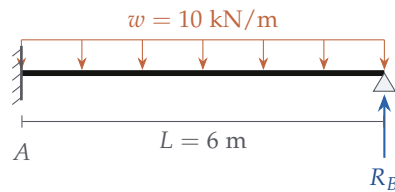
02 The force method

The force method, also called the flexibility or compatibility method, makes the beam determinate by removing one support reaction, the redundant. The released beam, the primary structure, now deflects where the real support allows no deflection. Superposing the deflection due to the loads and the deflection due to the redundant, and setting their sum to the real value, gives one equation per redundant. The method needs only the deflection formulas of determinate beams (tabulated in, for example, Gere and Goodno, *Mechanics of Materials*).

— *Release one reaction, then make the released beam fit its support again*

1. Count the redundants and pick one: here the prop force R_B .
2. Remove it. The primary structure is a cantilever, which is determinate.
3. Find the tip deflection under the load, $\delta_{B,w}$, and per unit R_B , f_{BB} .
4. Compatibility: the real prop does not move, so $\delta_{B,w} - R_B f_{BB} = 0$.
5. Equilibrium now gives every other reaction.

— The prop carries three eighths of the load: $R_B = 3wL/8$



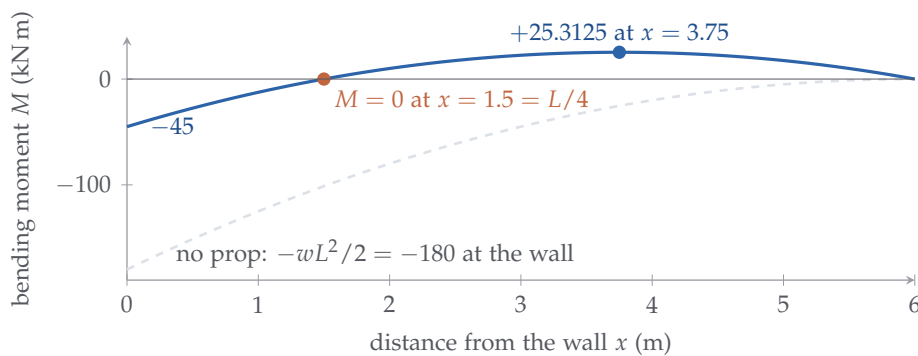
$$\delta_{B,w} = \frac{wL^4}{8EI} = 101.25 \text{ mm}$$

$$f_{BB} = \frac{L^3}{3EI}$$

$$R_B = \frac{\delta_{B,w}}{f_{BB}} = \frac{3wL}{8} = \mathbf{22.5 \text{ kN}}$$

Then $R_A = wL - R_B = 37.5 \text{ kN}$ and $M_A = wL^2/8 = 45 \text{ kN m}$, hogging.
 $EI = 16\,000 \text{ kN m}^2$ ($E = 200 \text{ GPa}$, $I = 8 \times 10^{-5} \text{ m}^4$); EI cancels in R_B .

— The prop cuts the peak moment from 180 to 45 kN m



Sagging positive: $M(x) = -45 + 37.5x - 10x^2/2 \text{ kN m}$.

03 Continuous beams

For a beam continuous over several supports, the convenient redundants are the bending moments over the interior supports. Cutting the beam into simply supported spans at those supports releases the continuity of slope; restoring it at each interior support gives Clapeyron's three-moment equation (1857), derived for example in Megson, *Structural and*

Stress Analysis. The form below assumes a uniform EI , supports at one level and a uniform load w_i on span i .

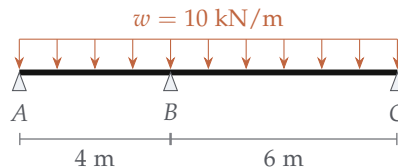
— The three-moment equation makes the slope continuous over each support

$$M_A L_1 + 2M_B(L_1 + L_2) + M_C L_2 = -\frac{w_1 L_1^3}{4} - \frac{w_2 L_2^3}{4}$$

- Redundants: the support moments. Each span becomes simply supported.
- Each right-hand term is $6EI$ times a simply supported end slope, $wL^3/24EI$.
- Pinned ends: $M_A = M_C = 0$. One interior support, one equation.

Uniform EI , supports at one level, sagging moments positive (Clapeyron, 1857).

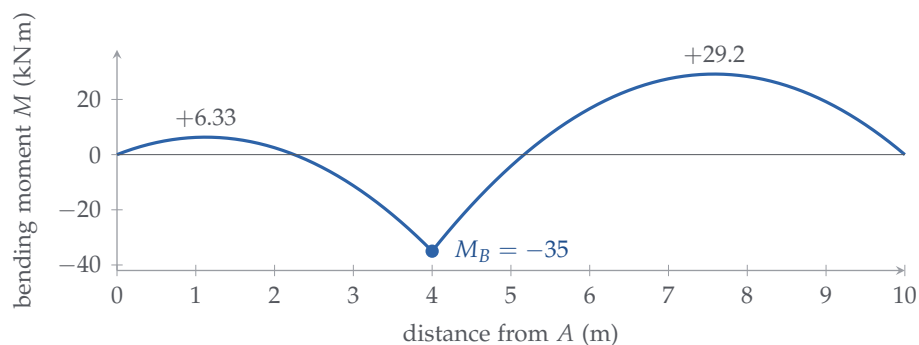
— Two spans of 4 and 6 m: the support moment is 35 kN m, hogging



$$\begin{aligned} 2M_B(4 + 6) &= -\frac{10 \cdot 4^3}{4} - \frac{10 \cdot 6^3}{4} \\ 20 M_B &= -160 - 540 = -700 \\ M_B &= -35 \text{ kN m} \end{aligned}$$

Span by span: $R_A = 11.25$, $R_B = 28.75 + 35.83 = 64.58$, $R_C = 24.17$ kN.

— The long span sags most; the support moment is the largest



Sagging positive. Peaks at zero shear: 1.125 m from A, 2.42 m from C.

04 Checks

Moment distribution (Cross, 1930) reaches the same support moment by a different route. Lock the interior joint against rotation, so each span starts with its fixed-end moments; with the far end pinned these are $wL^2/8$. Then release the joint: the unbalanced moment is

shared between the members in proportion to their stiffness, $3EI/L$ for a member whose far end is pinned. With only one joint and pinned far ends there is nothing to carry over, so a single balance is exact.

— *Moment distribution reaches the same 35 kN m in one balance*

At joint B	member BA	member BC
Stiffness $3EI/L$, share	0.6	0.4
Fixed-end moment $\pm wL^2/8$	+20	−45
Balance the −25 unbalance	+15	+10
Final end moment	+35	−35

End moments are clockwise-positive; ± 35 is the 35 kN m hogging moment found by the three-moment equation.

Far ends pinned: modified stiffness $3EI/L$, no carry-over (Cross, 1930).

— *Four checks catch most mistakes before they reach a drawing*

1. Vertical equilibrium: $11.25 + 64.58 + 24.17 = 100$ kN, and moments about A balance.
2. Limit case: equal spans must give $M_B = -wL^2/8$ (45 kN m for $L = 6$ m).
3. Compatibility: the deflection comes out zero at every support.
4. An independent solver: the site's beam visualizer, by the stiffness method, matches every number here.

Checks 3 and 4: beamdiag's exact-arithmetic reference solver, both examples.

— *What to remember*

1. Count first: the degree of indeterminacy is reactions minus three.
2. One compatibility equation per redundant: release it, then make the beam fit again.
3. For continuous beams, support moments and the three-moment equation are quickest.
4. Check with a second method and with compatibility, not only with equilibrium.